Tracer release from a stirred source tank

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Abstract

Release of an inert tracer from a stirred source tank into a conduit was measured.

1 Methods

The source tank held 250 L of water and was stirred at 120 rpm.

A dye tracer (5 g) was added at the start of each run (n = 4 runs).

2 Results

Tracer release reached 62 ± 5 % of the loaded dose after 30 min.

The release rate constant was approximately 0.04 1/min.

Between 85 and 95 % of the loaded dose had left the tank after 2 h.

Less than 2 % of the loaded dose remained in the tank at 6 h.

The outflow concentration peaked at 18.5 mg/L.

3 Limitations

Release was measured in a single tank geometry; other geometries were not tested.

Wall adsorption of the dye was not measured.